RAJHANSH YADAV

AI/ML Developer | Python | Machine Learning | BCA (AI/ML) Student
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SUMMARY

BCA (AI/ML) student at Noida International University who learns by building. Wrote a neural network from scratch in NumPy (98.1% test accuracy on handwritten digits, backpropagation verified with a gradient check) and a Python execution tracer that estimates time complexity automatically. Also built WhatsApp automation bots and a website for local businesses. Strong base in DSA, OOP (Java) and core CS, and looking for AI/ML internship opportunities.

SKILLS

Languages: Python, C, C++, Java, R
AI/ML & Data: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, PyTorch, TensorFlow, Keras, Hugging Face, LangChain, LlamaIndex
Core CS: Data Structures & Algorithms, OOP (Java), Operating Systems, Cyber Security
Web & Automation: Web Development, WhatsApp Bots
Tools: Git/GitHub, GitHub Actions (CI), pytest, VS Code, Jupyter Notebook, Google Colab, Cursor
Familiar With: Docker, Kubernetes, MLflow, Weights & Biases, Amazon SageMaker, Google Vertex AI, Microsoft Azure
Problem Solving: LeetCode: Python and maths problems (see GitHub)

PROJECTS

Neural Network From Scratch (NumPy) | GitHub

- Built a multi-layer perceptron (64→64→32→10) using only NumPy, with forward pass, backpropagation, Adam optimiser and L2 regularisation, to classify handwritten digits. Reached 98.1% test accuracy against 96.7% for a logistic-regression baseline.
- Verified backpropagation with a numerical gradient check (worst relative error 4.0e-07); avoided data leakage by scaling with train-only statistics and a stratified train/validation/test split. Added pytest tests and GitHub Actions CI, plus training curves, confusion matrix and error analysis.

Tech: Python, NumPy, scikit-learn, Matplotlib, pytest, GitHub Actions

Scope Visualizer: Python Execution Tracer & Complexity Estimator | GitHub

- Built a tool that records every function call (arguments, depth, time, memory) using sys.setprofile and tracemalloc, and writes one self-contained HTML report with a timeline, memory curve and collapsible call tree.
- Added an empirical complexity estimator that times a function at growing input sizes and fits O(1) to $O(n^3)$ models; it correctly identified O(n), O(n log n) and $O(n^2)$ for linear sum, merge sort and bubble sort. Tested with pytest and GitHub Actions CI.

Tech: Python, sys.setprofile, tracemalloc, HTML/SVG, pytest, GitHub Actions

NeuralFlow-AI — ML Pipeline for Breast-Tumor Classification | GitHub

- Built an end-to-end ML pipeline comparing Logistic Regression, Random Forest, and a small neural network via 5-fold cross-validation (ties broken toward the simpler model); the selected model reached 0.996 test ROC-AUC on unseen data.
- Tuned the decision threshold to catch 98%+ of malignant cases, verified the trade-off across 10 repeated train/test splits, and served predictions through a validated FastAPI endpoint; shipped with 17 pytest tests, Docker, and CI via GitHub Actions.

Tech: Python, scikit-learn, FastAPI, Pydantic, Docker, GitHub Actions, pytest

EXPERIENCE

Freelance Developer
Local food businesses

- Built WhatsApp automation bots for local food businesses to automate customer communication and support their business growth.
- Designed and developed a website for a local pizza shop.

EDUCATION

Bachelor of Computer Applications (AI/ML Specialisation) 2025 – 2028
Noida International University, Uttar Pradesh | Currently in 3rd semester

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification — DeepLearning.AI & Stanford Online (Coursera), Oct 2026
AI and Cybersecurity Awareness — TCS iON, "AI for All" initiative (2026)
Generative AI Essentials — TCS iON, "AI for All" initiative (2026)
Python — Kaggle (2026)
Python 101 for Data Science — IBM / Cognitive Class (2026)
Data Analysis with Python — freeCodeCamp, 300 hrs (2026)